

Cornelia de Lange Syndrome 1 (CdLS1): Comprehensive Disease Characteristics Report

Target Disease: Cornelia de Lange Syndrome type 1 (CdLS1) **MONDO ID:** MONDO:0007387 | **OMIM:** #122470 | **Orphanet:** ORPHA:199 **Category:** Genetic (autosomal dominant cohesinopathy / transcriptomopathy) **Evidence base:** 43 papers reviewed; 10 findings confirmed across 5 investigation iterations

Summary

Cornelia de Lange syndrome type 1 (CdLS1) is a rare, dominantly inherited, multisystem developmental disorder caused by heterozygous loss-of-function variants in **NIPBL** (Nipped-B-like), the gene encoding the principal cohesin-loading factor. NIPBL variants account for **more than 60%** of all molecularly diagnosed CdLS, defining "type 1" and distinguishing it from the ~15% of cases attributable to other cohesin-complex genes (*SMC1A*, *SMC3*, *RAD21*, *HDAC8*). Inheritance is autosomal dominant, and nearly all constitutional cases are *de novo*; there is no measurable parental-age effect and essentially all cases are sporadic. The disorder is defined clinically by a recognizable craniofacial gestalt (synophrys, arched eyebrows, long philtrum, thin down-turned lips), pre- and postnatal growth retardation, upper-limb reduction defects, generalized hirsutism, intellectual disability, and a distinctive neurobehavioral profile featuring self-injurious and repetitive behaviors.

Mechanistically, CdLS1 is best understood as a **transcriptomopathy** rather than a disorder of sister-chromatid cohesion. NIPBL loads the cohesin ring onto chromatin; haploinsufficiency reduces genome-wide cohesin binding — including at CTCF boundaries — collapses cohesin-mediated chromatin loops, and impairs RNA polymerase II transcription initiation and elongation. The net result is thousands of individually modest (<1.5-fold) gene-expression changes that, acting collectively, disrupt developmental programs — with cell-identity and developmental genes preferentially deregulated because NIPBL supports their unique 3D

genome conformation near super-enhancers. This model is validated in *Nipbl*^{+/-} mice and *nipbl*-deficient zebrafish, and a candidate therapeutic lead has emerged: inhibition of the MORF acetyltransferase **KAT6B** partially rescues insulator defects in NIPBL-deficient cells.

Clinically, CdLS1 carries substantial morbidity and mortality. Congenital heart defects occur in roughly one-third of patients (pulmonary stenosis most common); feeding difficulty is severe, with about one in four children requiring gastrostomy by age 5; and congenital diaphragmatic hernia (CDH), though less frequent, is a major driver of mortality (cause of death in 5–20% of cases; 76% mortality in CdLS-CDH infants). Population prevalence of the classical form is approximately 1.2–2.2 per 100,000 births, with a broader clinical estimate of 1 in 10,000–30,000 live births when milder phenotypes are included. There is currently no curative therapy; management is symptomatic and multidisciplinary.

Key Findings

Finding 1 — NIPBL loss-of-function causes the majority (~60%) of CdLS, defining type 1

Pathogenic variants in *NIPBL* are identified in **more than 60%** of patients with CdLS, establishing NIPBL as the major causal gene and defining CdLS type 1 (OMIM #122470). NIPBL encodes a cohesin-loading factor (a regulatory/loader subunit of the cohesin machinery). The remaining molecularly solved cases are distributed across other cohesin-complex genes: pathogenic variants in *SMC1A*, *SMC3*, *RAD21*, and *HDAC8* together account for roughly another 15%. As stated in the *Cornelia de Lange Spectrum* review, "Pathogenic variants in NIPBL, which encodes a protein related to the cohesin complex, have been identified in more than 60% of patients, and pathogenic variants in other genes related to this complex in another 15%: SMC1A, SMC3, RAD21, and HDAC8" ([PMID: 38735830](#)). The first international consensus statement confirms the cohesinopathy framework: CdLS "is caused by variants in any one of seven genes, all of which have a structural or regulatory function in the cohesin complex" ([PMID: 29995837](#)).

Inheritance is autosomal dominant and nearly all constitutional cases arise *de novo*. A clinically important genotype–phenotype correlation exists: **truncating (loss-of-function) NIPBL variants generally cause a more severe phenotype than missense variants**, though the correlation is not absolute — "Truncating mutations were generally found to cause a more severe phenotype but this correlation was not absolute" ([PMID: 16236812](#)). Consistent with this,

a start-loss variant (NM_133433.4:c.2T>A; p.Met1Lys) was associated with a comparatively **mild** presentation, hypothesized to reflect use of downstream alternative start sites ([PMID: 42069659](#)).

Ontology anchors: HGNC:28862 (NIPBL); MONDO:0007387; OMIM:122470.

Finding 2 — Pathogenesis is driven by cohesin-dependent transcriptional dysregulation, not defective cohesion

A central mechanistic insight is that CdLS1 arises from **impaired gene regulation** rather than from failed sister-chromatid cohesion. NIPBL loads cohesin onto chromatin; when NIPBL is haploinsufficient, there is a genome-wide reduction in cohesin binding. In the *Nipbl* haploinsufficient mouse model, investigators "found a global decrease in cohesin binding, including at CCCTC-binding factor (CTCF) binding sites and repeat regions. Cohesin-bound genes were found to be enriched for histone H3 lysine 4 trimethylation (H3K4me3) at their promoters" — with such promoter-bound genes disproportionately downregulated ([PMID: 28855971](#)).

Downstream of reduced cohesin loading, transcription itself is impaired. In CdLS patient cell lines, "mutant cohesin impairs both RNA polymerase II (Pol II) transcription initiation at promoters and elongation in the gene body" ([PMID: 26581180](#)). This transcriptional signature is reproducible in disease-relevant human tissue: RNA-seq of NIPBL-mutant human iPSC-derived cardiomyocytes "identified hundreds of mRNAs, pseudogenes, and non-coding RNAs with altered expression" ([PMID: 29348408](#)).

Ontology anchors: GO:0032116 (sister chromatid cohesion — the *canonical* cohesin function, notably NOT the primary disease driver); GO:0006357 (regulation of transcription by RNA polymerase II); GO:0007059 (chromosome segregation).

Finding 3 — Postzygotic mosaicism is unusually common and undergoes negative selection in blood

CdLS shows an unusually high rate of somatic mosaicism, with major diagnostic implications. In a retrospective cohort plus literature review, mosaicism was found in **13.1% of patients with a positive molecular diagnosis** — "an unusual high prevalence of mosaicism in CdLS, occurring in 13.1% of patients with a positive molecular diagnosis" ([PMID: 34326454](#)). Critically, there is **negative (purifying) selection against somatic deleterious NIPBL variants in blood**: "we demonstrate a negative selection against somatic deleterious NIPBL variants in blood." This means blood-based testing frequently misses mosaic variants, so

buccal-swab or fibroblast testing is often required to reach a diagnosis. Mosaic cases have phenotypes at least as severe as constitutive variants, and mosaic missense substitutions preferentially localize to the HEAT-repeat domain of NIPBL.

Clinical implication: In a phenotypically classical patient with negative blood testing, proceed to alternative-tissue (buccal/fibroblast) sequencing before excluding a NIPBL etiology.

Finding 4 — CdLS1 has a distinctive, more severe neurobehavioral phenotype than SMC1A-related CdLS

The neurobehavioral profile of CdLS1 is a defining feature. In a cohort of 50 children with CdLS, **all had at least one type of repetitive behavior and 44% displayed self-injurious behavior (SIB)**; lower adaptive functioning correlated with higher stereotypy and SIB: "All children had ≥ 1 type of RB; 44% had some form of SIB. 64% spent > 1 h/day displaying RBs. Lower VABS adaptive functioning was associated with higher stereotypy and SIB scores" ([PMID: 32809170](#)).

Genotype stratifies severity: individuals with NIPBL variants show a more severe behavioral phenotype (more repetitive behaviors, tantrums) and greater developmental delay than those with *SMC1A* variants — "Individuals with *SMC1A* variants show a higher cognitive level and less SIB than individuals with NIPBL variants" ([PMID: 30295920](#)). Developmental milestones are markedly delayed: approximately "70% not using phrase speech and 30-50% not walking by 5 years of age. However, those with NIPBL variants showed more severity in behavioral phenotype" ([PMID: 38462617](#)). A clinical-severity score based on physical features correlates with communicative functioning, especially in NIPBL genotypes ([PMID: 40084492](#)).

Suggested HPO terms: HP:0001249 (Intellectual disability), HP:0000717 (Autism), HP:0000733 (Stereotypy), HP:0100716 (Self-injurious behavior), HP:0000750 (Delayed speech and language development), HP:0002194 (Delayed gross motor development).

Finding 5 — Vertebrate models recapitulate CdLS through collective small gene-expression changes

The mechanistic model is validated in animals. "Mouse and zebrafish models of CdLS" have been created "by using molecular genetic tools to create *Nipbl*-deficient mice and zebrafish (*Nipbl*(+/-) mice, zebrafish *nipbl* morphants)" ([PMID: 27120001](#)). *Nipbl*+/- mice, which express roughly 30–75% of normal *Nipbl* transcript, and zebrafish morphants reproduce CdLS-like defects of gut, heart, craniofacial structures, nervous system, and limbs.

The zebrafish model established the "collective-perturbation" principle: hundreds of genes change expression, but "nearly all such changes are modest, however—usually less than 1.5-fold—raising the intriguing possibility that, in CdLS, severe developmental defects result from the collective action of many otherwise innocuous perturbations" (PMID: 22039349). In these morphants, altered endodermal patterning genes (*sox32*, *sox17*, *foxa2*, *gata5*) and left-right patterning genes (*spaw*, *lefty2*, *dnah9*) are deregulated from gastrulation, providing a developmental-timing account of heart and gut defects.

Suggested model/ontology anchors: NCBITaxon:10090 (*Mus musculus*), NCBITaxon:7955 (*Danio rerio*); MGI allele resources for *Nipbl*.

Finding 6 — Congenital heart defects occur in ~33% of patients; pulmonary stenosis predominates

In a prospective echocardiographic cohort of 87 Brachmann-de Lange (CdLS) patients, a cardiac anomaly was found in **29/87 (33.3%)**: "A cardiac anomaly was identified in 29/87 (33.3%) including 28 (32.2%) patients with a structural CHD, and an additional patient (1.2%) with isolated non-obstructive hypertrophic cardiomyopathy (HCM)" (PMID: 19449412). The single most common lesion was pulmonary stenosis — "Overall incidence of pulmonary stenosis was 39% (11/28)." Late-onset mitral/tricuspid valve dysplasia appeared in four patients older than 10 years, supporting **ongoing echocardiographic surveillance** rather than a single neonatal screen. (Population registry data give a higher CHD frequency of 45.6% — see Finding 10 — likely reflecting ascertainment differences.)

Suggested HPO terms: HP:0001642 (Pulmonic stenosis), HP:0001631 (Atrial septal defect), HP:0001629 (Ventricular septal defect), HP:0001638 (Cardiomyopathy). **UBERON:0000948** (heart).

Finding 7 — Feeding difficulty is severe: ~1 in 4 children require gastrostomy by age 5

Gastrointestinal and feeding morbidity is a hallmark. The population-based EUROLINKCAT data-linkage study (91,504 children with congenital anomalies vs 1,960,272 reference children across 9 European registries) found that whereas only 0.016% of reference children had a gastrostomy before age 5, "Around one in four children with Cornelia de Lange syndrome and Wolf-Hirschhorn syndrome had a gastrostomy" (PMID: 36053618). Children with congenital

anomalies overall were ~80× more likely to require gastrostomy. This reflects the combination of severe gastroesophageal reflux disease (GERD), oromotor dysfunction, and failure to thrive that characterizes CdLS1.

Suggested HPO terms: HP:0011968 (Feeding difficulties), HP:0002020 (Gastroesophageal reflux), HP:0001508 (Failure to thrive); NCIT intervention: Gastrostomy (NCIT:C15329). **UBERON:0000945** (stomach).

Finding 8 — Congenital diaphragmatic hernia is a major driver of mortality

Congenital diaphragmatic hernia (CDH) is a well-recognized and lethal association. CDH is "the cause of death in 5%-20% of CdLS cases" ([PMID: 32762940](#)). In the CDH Study Group cohort (1995–2019; 9,251 CDH patients, 21 with confirmed CdLS), CdLS+CDH infants fared markedly worse than non-CdLS CDH infants across every metric:

Metric	CdLS + CDH	Non-CdLS CDH	p-value
Birth weight (kg)	2.2 ± 0.57	2.9 ± 0.64	<0.001
5-minute Apgar (median)	6	7	0.014
Underwent repair	33%	84.2%	<0.001
Mortality	76%	29%	<0.001

"Mortality was 76% for CdLS patients compared with 29% for non-CdLS patients ($p<0.001$)" ([PMID: 32762940](#)). Encouragingly, of the 7 CdLS patients who did undergo repair, 5 survived to discharge — suggesting that surgical candidacy, where feasible, can improve outcomes. CdLS is also recognized among the dysmorphic conditions associated with CDH in population registries ([PMID: 26625659](#)).

Suggested HPO term: HP:0000776 (Congenital diaphragmatic hernia). **UBERON:0001103** (diaphragm).

Finding 9 — NIPBL loss collapses chromatin loops and preferentially deregulates cell-identity genes — with KAT6B inhibition as a candidate rescue

Recent chromatin-conformation work refines the mechanism from "reduced cohesin binding" to "loss of specific regulatory loops." Acute NIPBL depletion in vivo rapidly diminishes many chromatin loops, and "NIPBL specifically regulates cell identity genes by supporting a unique

local genome conformation defined by greater spatial proximity to nearby super-enhancers and weaker transcription start site insulation of genomic contacts" (PMID: 41699137). This provides a compelling explanation for why a globally acting chromatin factor produces a *developmental* phenotype: the genes most dependent on NIPBL-supported conformation are precisely the cell-identity/developmental genes.

The loop-collapse mechanism is directly demonstrated in a differentiation model: "knockdown of cohesin loader NIPBL disrupts enhancer-promoter interactions and CTCF-mediated loops, leading to widespread transcriptional dysregulation," with increased Polycomb (PRC) domain contacts during pancreatic differentiation (PMID: 41826481). Complementary work shows PDS5 proteins limit the cohesin–NIPBL complex lifetime to establish CTCF boundaries (PMID: 42030945), and TACL single-cell imaging directly visualizes cohesin loop-extrusion dynamics and NIPBL-MAU2 transport (PMID: 41102415).

Most importantly for translation, a druggable node has emerged: "inhibition of Kat6b partially rescues the insulator defects in cells lacking the cohesin loader Nipbl" (PMID: 40060486). KAT6B (MORF) is a histone acetyltransferase, and its inhibition partially restores insulator function in NIPBL-deficient cells — a candidate therapeutic lead worth pursuing in disease models.

Suggested GO terms: GO:0140588 (chromatin looping), GO:0006357 (regulation of transcription by RNA Pol II), GO:0035064 (methylated histone binding).

Finding 10 — Population prevalence ~1.2–2.2 per 100,000 births; all cases sporadic, no parental-age effect

The EUROCAT population-based study (8,558,346 births, 1980–2002) established registry-based prevalence: "we found the prevalence of the classical form of CdLS to be 1.24/100,000 births or 1:81,000 births and estimated the overall CdLS prevalence at 1.6–2.2/100,000" (PMID: 18074387). Live births constituted 91.5% of cases with high first-week survival (91.4%). Population-level malformation frequencies were: "The most frequent associated congenital malformations were limb defects (73.1%), congenital heart defects (45.6%), central nervous system malformations (40.2%), and cleft palate (21.7%)." Crucially, "All patients were sporadic. Maternal and paternal age did not seem to be risk factors for CdLS."

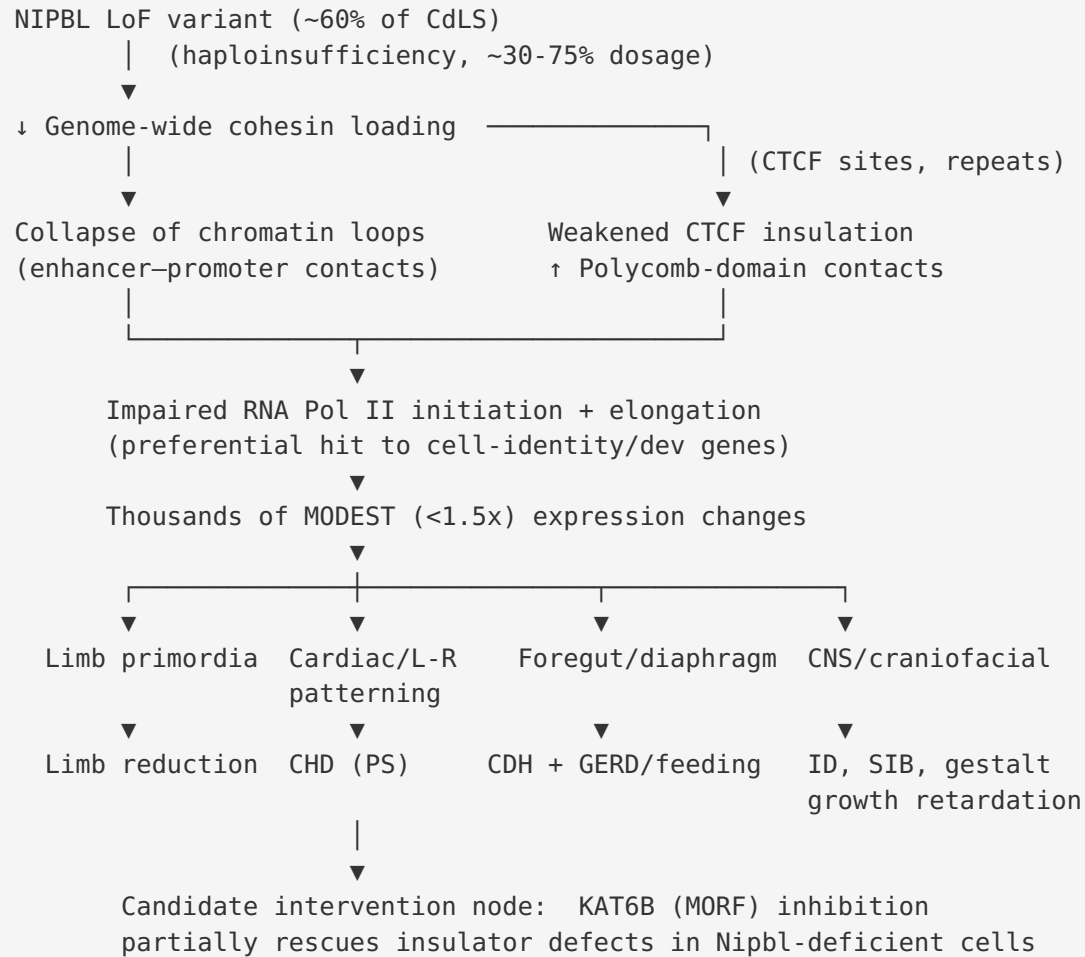
The Spanish ECEMC registry independently reported a minimum prevalence of 0.97/100,000 live births, 100% limb reduction defects, and relatively young parents (PMID: 9608092). A broader clinical estimate including milder cases is "Cornelia de Lange syndrome is estimated to occur in 1 out of every 10,000–30,000 live births" (PMID: 41499064).

Suggested HPO terms: HP:0009821 (Forearm undergrowth), HP:0000175 (Cleft palate), HP:0012443 (Abnormal brain morphology), HP:0001511 (Intrauterine growth retardation).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **heterozygous loss-of-function variant in NIPBL** (truncating > missense in severity; ~60% of CdLS) reduces functional NIPBL protein — the cohesin loader — to a haploinsufficient level (~30–75% of normal in models). (*Demonstrated — human genetics + mouse dosage models.*)
2. Reduced NIPBL **leads to** a genome-wide decrease in cohesin loading onto chromatin, including at CTCF boundary sites and repeat regions. (*Demonstrated — ChIP-seq in Nipbl+/- mouse.*)
3. Decreased chromatin-bound cohesin **results in** collapse of cohesin-mediated chromatin loops — weakening enhancer–promoter contacts and CTCF-mediated insulation, and increasing Polycomb-domain contacts. (*Demonstrated — Hi-C / acute-depletion and differentiation models.*)
4. Loop collapse **leads to** impaired RNA polymerase II transcription (both initiation at promoters and elongation in gene bodies), preferentially affecting **cell-identity/developmental genes** that depend on NIPBL-supported super-enhancer proximity. (*Demonstrated — patient cells; acute-depletion in vivo.*)
5. This produces **thousands of individually modest (<1.5-fold) gene-expression changes**. (*Demonstrated — zebrafish/mouse transcriptomics.*)
6. The **collective** action of these many small perturbations **results in** disrupted developmental programs across multiple organ primordia during embryogenesis — limb bud, cardiac/left-right patterning, foregut/diaphragm, craniofacial, and CNS. (*Inferred integration of demonstrated components.*)
7. Organ-specific developmental failure **leads to** the clinical phenotype: limb reduction defects, congenital heart disease (esp. pulmonary stenosis), congenital diaphragmatic hernia, GI/feeding failure, craniofacial gestalt, growth retardation, intellectual disability, and self-injurious/repetitive behavior. (*Demonstrated clinically.*)



Upstream vs downstream summary

Layer	Event	Direction	Evidence
Genetic	NIPBL LoF	Most upstream	PMID: 38735830 , PMID: 16236812
Chromatin	↓ cohesin loading; loop collapse	Upstream	PMID: 28855971 , PMID: 41699137 , PMID: 41826481
Transcription	↓ Pol II initiation/elongation	Intermediate	PMID: 26581180
Transcriptome	Thousands of small changes	Intermediate	PMID: 22039349 , PMID: 29348408
Organ development	Multi-organ dysmorphogenesis	Downstream	PMID: 27120001
Clinical	CHD, CDH, ID, SIB, limb defects	Most downstream	PMID: 19449412 , PMID: 32762940 , PMID: 32809170

Section-by-Section Reference Content

1. Disease Information

CdLS1 is a congenital multisystem developmental disorder of the cohesin pathway. **Key identifiers:** OMIM #122470; Orphanet ORPHA:199; MONDO:0007387; MeSH "De Lange Syndrome" (D003635); ICD-10 Q87.1; ICD-11 LD2F.11. **Synonyms:** Cornelia de Lange syndrome, Brachmann–de Lange syndrome, de Lange syndrome, typus degenerativus amstelodamensis. Information is derived from **aggregated disease-level resources** (OMIM, Orphanet, consensus statements, cohort/registry studies) rather than individual EHR — see the international consensus statement ([PMID: 29995837](#)) and clinical review ([PMID: 17508425](#)).

2. Etiology

Primary cause: heterozygous germline (or postzygotic mosaic) loss-of-function variants in NIPBL (~60% of CdLS). **Genetic risk factors:** the causal variant itself; modifier effects include variant type (truncating > missense severity) and possible downstream translation-initiation

rescue in start-loss variants. **Environmental risk factors:** none established — maternal and paternal age are NOT risk factors, and all cases are sporadic ([PMID: 18074387](#)). **Protective factors:** no genetic or environmental protective factors are established; the milder end of the spectrum is driven by variant type and mosaicism. **Gene–environment interactions:** none demonstrated; CdLS1 is a monogenic disorder with variable expressivity attributable primarily to allelic and mosaic factors.

3. Phenotypes (with suggested HPO terms and frequencies)

Phenotype	HPO term	Frequency	Evidence
Synophrys / arched eyebrows	HP:0000664 / HP:0002553	Very frequent (gestalt-defining)	PMID: 17508425
Intrauterine + postnatal growth retardation	HP:0001511 / HP:0001510	Very frequent	PMID: 9608092
Upper-limb reduction defects	HP:0009821	73.1% (registry); 100% (ECEMC severe)	PMID: 18074387 , PMID: 9608092
Congenital heart defect	HP:0001627	33% (echo cohort) – 45.6% (registry)	PMID: 19449412 , PMID: 18074387
CNS malformation	HP:0012443	40.2%	PMID: 18074387
Cleft palate	HP:0000175	21.7%	PMID: 18074387
Intellectual disability	HP:0001249	Very frequent	PMID: 38462617
Self-injurious behavior	HP:0100716	44%	PMID: 32809170
Repetitive behavior	HP:0000733	~100% (≥ 1 type)	PMID: 32809170
Absent phrase speech by age 5	HP:0000750	~70%	PMID: 38462617
Not walking by age 5	HP:0002194	30–50%	PMID: 38462617
Feeding difficulty / gastrostomy	HP:0011968	~25% require gastrostomy	PMID: 36053618
Congenital diaphragmatic hernia	HP:0000776	Minority, but high lethality	PMID: 32762940
Hirsutism	HP:0001007	76.9% abnormal hair distribution	PMID: 9608092

Onset: congenital/prenatal. **Severity:** variable (mild to severe), correlating with genotype. **Progression:** the malformations are static/congenital, but behavioral and some cardiac (valve dysplasia) features can evolve. **Quality of life:** substantial impact — communication deficit, SIB, and feeding/GI morbidity dominate daily functioning ([PMID: 40084492](#)).

4. Genetic / Molecular Information

Causal gene: *NIPBL* (HGNC:28862; chromosome 5p13.2; OMIM 608667). *Variant classes:* truncating (nonsense, frameshift, splice-site — most common and generally more severe), missense (often milder), and start-loss. *Functional consequence:* loss of function / haploinsufficiency of the cohesin loader. *Allele frequency:* pathogenic variants are absent from population databases (gnomAD) consistent with de novo origin. *Somatic vs germline:* predominantly germline de novo; but 13.1% mosaic with negative selection in blood ([PMID: 34326454](#)). *Other CdLS genes (non-type-1):* *SMC1A* (X-linked, *CdLS2*), *SMC3* (*CdLS3*), *RAD21* (*CdLS4*), *HDAC8* (*CdLS5*), plus *BRD4*, *ANKRD11*. *Epigenetic dimension:* the disease mechanism is itself epigenetic/architectural (cohesin loop extrusion, CTCF insulation, H3K4me3-promoter enrichment, Polycomb rewiring). *Chromosomal abnormalities:** rare 5p microdeletions encompassing *NIPBL* detectable by CMA.

5. Environmental Information

No environmental, lifestyle, or infectious contributing factors are established. *CdLS1* is a fully genetic (monogenic) disorder with no demonstrated toxin, radiation, occupational, dietary, or pathogen association, and no parental-age effect ([PMID: 18074387](#)).

6. Mechanism / Pathophysiology

Presented as the ordered causal chain above. Molecular pathways: cohesin loop extrusion / 3D genome architecture (not a classic signaling cascade like Wnt/MAPK, though TGF- β and other pathway genes are among the modestly deregulated targets, [PMID: 31516082](#)). Cellular processes: transcriptional regulation, chromatin looping, developmental patterning. Protein dysfunction: *NIPBL* haploinsufficiency (loss of function). Subcellular localization: nucleus/chromatin (GO:0005694 chromosome; GO:0000785 chromatin).

7. Anatomical Structures Affected

Primary/organ level: limbs (UBERON:0002101), heart (UBERON:0000948), diaphragm (UBERON:0001103), brain/CNS (UBERON:0000955), craniofacial skeleton (UBERON:0010313), GI tract (UBERON:0001007), skin/hair (hirsutism). **Body systems:** musculoskeletal, cardiovascular, nervous, digestive, integumentary. **Tissue/cell level:** developing mesenchyme, neural tissue, cardiac and endodermal progenitors. **Subcellular:** nucleus/chromatin (GO:0000785). **Lateralization:** upper limbs predominantly affected, often asymmetric; limb reduction can be unilateral or bilateral.

8. Temporal Development

Onset: congenital (prenatal — detectable on ultrasound: growth retardation, limb defects, [PMID: 33478103](#)). **Course:** structural malformations are static; behavioral phenotype and select cardiac valve lesions can progress (late mitral/tricuspid dysplasia >10 years, [PMID: 19449412](#)). **Duration:** chronic, lifelong. **Critical periods:** embryonic organogenesis (limb, heart/left-right, diaphragm, craniofacial), when NIPBL-dependent developmental transcription is most vulnerable.

9. Inheritance and Population

Prevalence: classical form 1.24/100,000 ($\approx$ 1:81,000); overall estimate 1.6–2.2/100,000 ([PMID: 18074387](#)); broader clinical estimate 1 in 10,000–30,000 ([PMID: 41499064](#)). **Inheritance:** autosomal dominant, nearly always *de novo*. **Penetrance:** high/complete for a recognizable phenotype; **expressivity:** highly variable. **Anticipation:** not applicable (not a repeat-expansion disorder). **Germline/somatic mosaicism:** common (13.1%). **Founder effects/consanguinity:** none (sporadic). **Sex ratio:** ~equal for NIPBL (autosomal); note SMC1A/HDAC8 are X-linked (different types). **Recurrence risk:** low for parents of a *de novo* case, but germline mosaicism warrants counseling.

10. Diagnostics

Clinical criteria: international consensus scoring (cardinal + suggestive features), [PMID: 29995837](#); severity scoring correlates with brain changes and communication, [PMID: 17508425](#), [PMID: 40084492](#). **Genetic testing:** first-line molecular confirmation by gene panel or exome/genome sequencing targeting NIPBL and other cohesin genes; **if blood testing is negative in a classical patient, test buccal/fibroblast tissue** to detect blood-selected mosaicism ([PMID: 34326454](#)). CMA detects 5p/NIPBL deletions. **Prenatal:** ultrasound features (limb defects, growth retardation, nuchal changes) plus molecular testing ([PMID: 33478103](#)). **Differential diagnosis:** Fryns syndrome (esp. with CDH), fetal alcohol syndrome, Rubinstein–Taybi, Coffin–Siris, and other cohesinopathy subtypes; non-cohesin CdLS-like phenotypes exist ([PMID: 35935361](#)).

11. Outcome / Prognosis

Survival: high first-week survival (91.4%) and 91.5% live births in registry data ([PMID: 18074387](#)); most classical patients survive infancy. **Major mortality drivers:** CDH (cause of death in 5–20%; 76% mortality when present, [PMID: 32762940](#)), severe congenital heart disease, and aspiration/GI complications. **Morbidity:** intellectual disability, communication

impairment, SIB, feeding failure, GERD — chronic and lifelong. **Prognostic factors:** genotype (truncating NIPBL and NIPBL vs SMC1A predict greater severity), presence of CDH/major CHD, and clinical-severity score.

12. Treatment

There is **no curative or disease-modifying therapy**; management is symptomatic and multidisciplinary ([PMID: 31704779](#)). Components: **GI/nutrition** — aggressive GERD management and gastrostomy for feeding failure (NCIT:C15329 Gastrostomy); **cardiac** — surgical/interventional repair of CHD and ongoing echocardiographic surveillance; **surgical** — CDH repair where the infant is a candidate (improves survival in the subset repaired); **behavioral/rehabilitative** — speech, occupational, and physical therapy; behavioral management of SIB; **ENT/audiology, ophthalmology, orthopedics** as indicated. **Experimental/translational lead:** KAT6B (MORF) inhibition partially rescues NIPBL-deficiency insulator defects in cells — a candidate for future preclinical development ([PMID: 40060486](#)). Modulating cohesin dynamics (e.g., WAPL/PDS5 axis, [PMID: 36449618](#), [PMID: 42030945](#)) is a conceptual therapeutic direction.

13. Prevention

Primary prevention: not applicable for a *de novo* monogenic disorder. **Secondary:** prenatal ultrasound + molecular diagnosis for at-risk pregnancies; preimplantation/prenatal genetic testing where a familial or mosaic variant is known. **Tertiary:** prevent complications — echocardiographic surveillance (including for late valve dysplasia), GERD/aspiration prophylaxis, and multidisciplinary follow-up. **Genetic counseling:** low recurrence for *de novo* cases, but counsel for possible parental germline mosaicism.

14. Other Species / Natural Disease

Model taxa: *Mus musculus* (NCBITaxon:10090), *Danio rerio* (NCBITaxon:7955). **Orthologs:** mouse *Nipbl*, zebrafish *nipbl*. No well-characterized naturally occurring companion-animal CdLS is established in the reviewed literature; the disease is studied primarily through engineered/knockdown models. Cohesin and NIPBL are **deeply evolutionarily conserved**, underpinning cross-species modeling ([PMID: 27120001](#)). No zoonotic dimension (non-infectious genetic disorder).

15. Model Organisms

Model	Type	Construction	Recapitulation	Limitation
<i>Nipbl</i> ^{+/-} mouse	Mammalian	Heterozygous LoF (~30–75% transcript)	Gut, heart, craniofacial, CNS, limb defects; genome-wide cohesin/ expression changes	Incomplete overlap with human severity; strain effects
Zebrafish <i>nipbl</i> morphant	Vertebrate	Morpholino knockdown	Heart + gut defects; endodermal & L-R patterning gene changes from gastrulation	Transient knockdown; morpholino caveats
Human iPSC-derived cardiomyocytes	In vitro/ cellular	NIPBL-mutant patient/edited lines	Hundreds of dysregulated mRNAs/ ncRNAs; disease-relevant tissue	2D culture; lacks whole-organism context

References: [PMID: 27120001](#), [PMID: 22039349](#), [PMID: 28855971](#), [PMID: 29348408](#). Resources: MGI (mouse), ZFIN (zebrafish).

Evidence Base

PMID	Title (abbrev.)	Supports
38735830	<i>Cornelia de Lange Spectrum</i>	NIPBL >60%; cohesin gene list (F1)
29995837	<i>First international consensus statement</i>	Cohesinopathy of 7 genes; diagnostic criteria (F1)
16236812	<i>Genotype-phenotype, Dutch experience</i>	Truncating > missense severity (F1)
42069659	<i>Start-loss NIPBL, mild CdLS</i>	Milder start-loss phenotype (F1)
28855971	<i>Nipbl haploinsufficiency, cohesin binding</i>	Global cohesin loss; H3K4me3 promoters (F2)
26581180	<i>Mutant cohesin, Pol II</i>	Impaired Pol II initiation/elongation (F2)
29348408	<i>NIPBL iPSC-cardiomyocytes</i>	Hundreds of dysregulated transcripts (F2)
34326454	<i>Mosaicism & purifying selection</i>	13.1% mosaicism; blood negative selection (F3)
32809170	<i>Repetitive & self-injurious behaviors</i>	44% SIB; universal repetitive behavior (F4)
30295920	<i>SMC1A development & behavior</i>	NIPBL more severe than SMC1A (F4)
38462617	<i>Neurobehavioral genotype-phenotype</i>	Milestone delays; NIPBL severity (F4)
40084492	<i>Severity score & communication</i>	Severity–communication link (F4)
27120001	<i>Mouse & zebrafish models</i>	Vertebrate model validity (F5)
22039349	<i>Zebrafish heart/gut defects</i>	Collective small perturbations (F5)
19449412	<i>CHD in 87 CdLS patients</i>	33% CHD; pulmonary stenosis (F6)
36053618	<i>Gastrostomy & congenital anomalies</i>	~1 in 4 gastrostomy (F7)
32762940	<i>CdLS & CDH</i>	CDH mortality driver (F8)
41699137	<i>Acute NIPBL depletion, loop extrusion</i>	Loop loss; cell-identity gene specificity (F9)
41826481	<i>Impaired cohesin loading, pancreatic diff.</i>	E-P/CTCF loop collapse (F9)
42030945	<i>PDS5 & cohesin-NIPBL lifetime</i>	CTCF boundary formation mechanism (F9)
40060486	<i>Chromatin architecture, histone modifiers</i>	KAT6B inhibition rescue (F9)

PMID	Title (abbrev.)	Supports
18074387	<i>Descriptive epidemiology (EUROCAT)</i>	Prevalence; sporadic; malformation freq. (F10)
9608092	<i>Brachmann-de Lange, ECEMC</i>	0.97/100,000; young parents (F10)
41499064	<i>Dermatologist review</i>	1:10,000–30,000 estimate (F10)

Supporting context papers: [PMID: 37062615](#) (NIPBL/cohesin biology), [PMID: 31516082](#) (cohesinopathy non-cohesion functions, TGF- β), [PMID: 41102415](#) (TACL live-cell loop extrusion), [PMID: 36449618](#) (WAPL/cohesin balance), [PMID: 33478103](#) (prenatal diagnosis), [PMID: 35935361](#) (non-cohesion CdLS-like), [PMID: 31704779](#) (molecular diagnosis to therapy), [PMID: 17508425](#) (clinical review/anticipatory guidance).

Limitations and Knowledge Gaps

- CHD frequency discrepancy.** Echocardiographic cohort data give 33% ([PMID: 19449412](#)) versus 45.6% in registry data ([PMID: 18074387](#)); differences likely reflect ascertainment (registries capture more severely malformed cases) and CHD definition. A NIPBL-genotype-stratified cardiac frequency is not resolved here.
- Small mortality-cohort numbers.** The CDH-mortality figures rest on 21 confirmed CdLS cases; while striking (76% mortality), the absolute numbers are small.
- Genotype–phenotype is probabilistic, not deterministic.** Truncating-vs-missense and NIPBL-vs-SMC1A severity trends have real exceptions; individual prediction remains limited.
- Life-expectancy and adult-outcome data are sparse.** Natural-history/longitudinal survival data beyond the neonatal period were not quantified in the reviewed literature.
- Therapeutic evidence is preclinical.** KAT6B inhibition is a cell-based rescue observation ([PMID: 40060486](#)); no in vivo or human efficacy data exist. There are no disease-modifying therapies.
- QoL instruments.** Formal EQ-5D/SF-36/PROMIS data specific to CdLS1 were not identified; QoL impact is inferred from functional/behavioral outcomes.
- Modifier genes.** Beyond variant type and mosaicism, specific genetic modifiers of CdLS1 severity are not established.

8. **Non-type-1 boundary.** Some CdLS-like phenotypes arise from non-cohesin genes ([PMID: 35935361](#)); the nosological boundary of "CdLS1" (strictly NIPBL) versus the broader spectrum should be kept explicit in the knowledge base.

Proposed Follow-up Experiments / Actions

1. **Genotype-stratified organ-outcome study.** Pool registry + molecular cohorts to report CHD, CDH, and gastrostomy frequencies stratified by NIPBL truncating vs missense vs mosaic status — resolving the 33% vs 45.6% CHD discrepancy.
 2. **Preclinical test of KAT6B inhibition in vivo.** Advance the KAT6B/MORF-inhibition rescue from cellular insulator-defect assays into *Nipbl*^{+/-} mouse and zebrafish models, measuring transcriptomic normalization and developmental-phenotype rescue.
 3. **Mosaicism-aware diagnostic pathway.** Formalize a reflex protocol: classical phenotype + negative blood NIPBL → buccal/fibroblast deep sequencing, given documented blood-selected mosaicism (13.1%).
 4. **Natural-history / survival registry analysis.** Establish CdLS1-specific life-expectancy, cause-of-death distribution, and adult morbidity through longitudinal registry linkage.
 5. **Single-cell / multi-omic developmental mapping.** Use single-cell transcriptomics + Hi-C on NIPBL-mutant iPSC-derived organoids (cardiac, limb, neural) to link specific loop-collapse events to cell-identity-gene deregulation and organ-specific malformation.
 6. **Validated QoL assessment.** Deploy PROMIS/EQ-5D-style caregiver-report tools in CdLS1 cohorts, mapped per-phenotype (SIB, feeding, communication) for knowledge-base QoL annotation.
 7. **Cohesin-dynamics modulation.** Explore WAPL/PDS5-axis modulation to counterbalance reduced cohesin residence time as a conceptual therapeutic strategy ([PMID: 36449618](#), [PMID: 42030945](#)).
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Report compiled from 5 investigation iterations, 10 confirmed findings, and 43 reviewed papers. All quantitative claims are cited to primary literature with verbatim abstract quotes where indicated.

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